

brown open silicon

designing open chips. together.

agenda



icebreaker

manufacturing
process

transistor
architecture,
sizing

stick diagrams, reading DRC &
LVS,
reverse engineering, &
tool introduction



fixing LVS

icebreaker



which spongebob gif are you today?

manufacturing process



quick videos on the silicon manufacturing

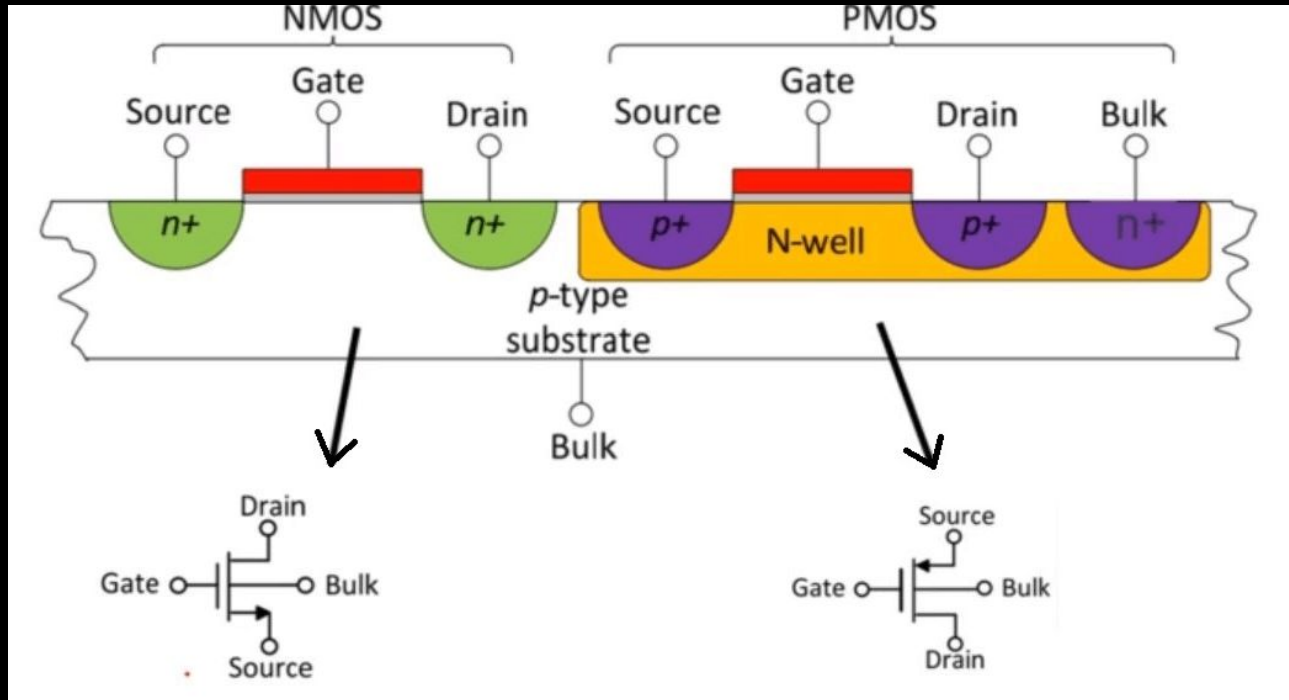
[Ted ED how microchips are made](#)

[ASML's microchips](#)

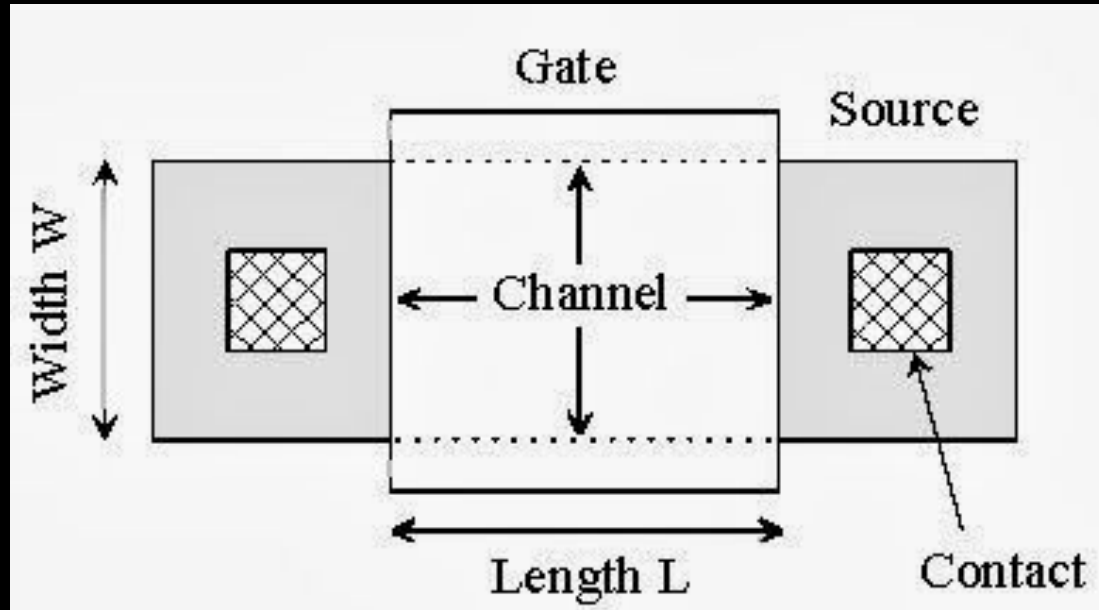


**who is able to draw a transistor
architecture from memory?**

transistor architecture



transistor architecture

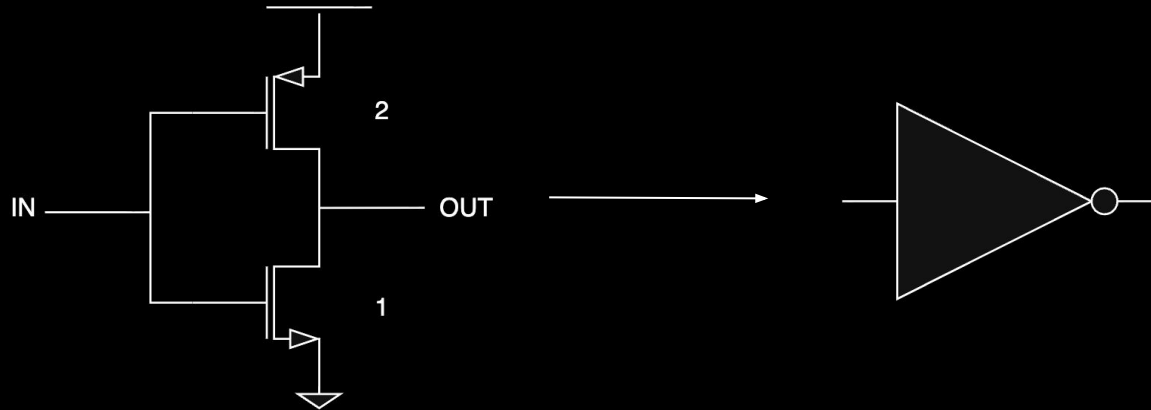


transistor sizing

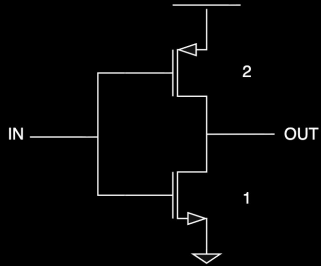


typically changing W , keeping L at a minimum

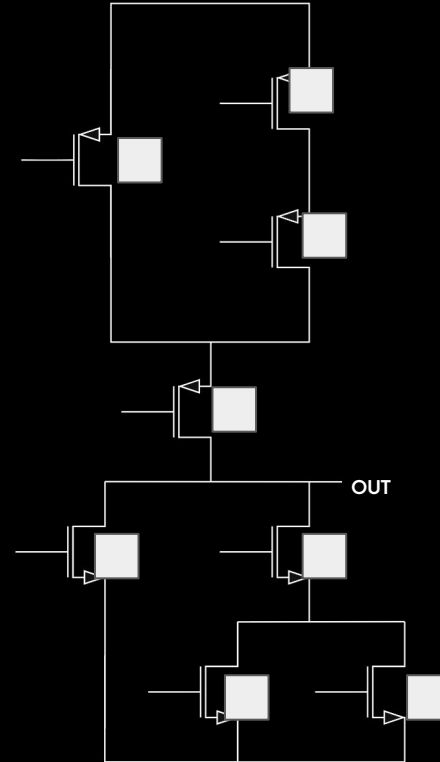
general sizing: can base it off the inverter



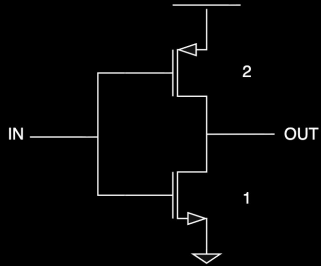
transistor sizing



1. start with the shortest path to output (nmos or pmos)
2. use the inverter as reference
3. scale!

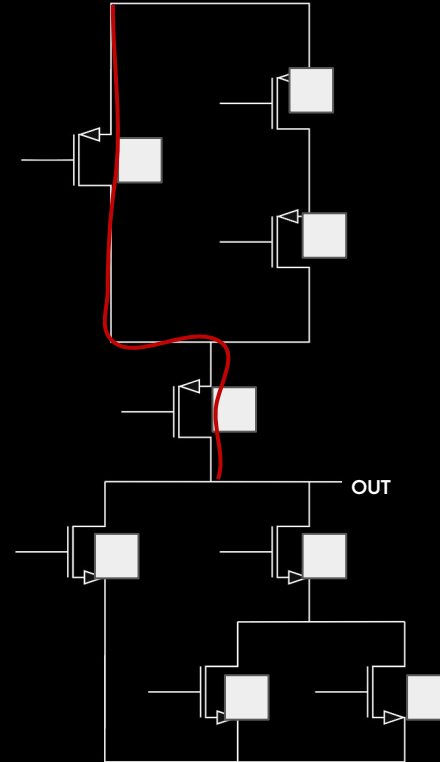


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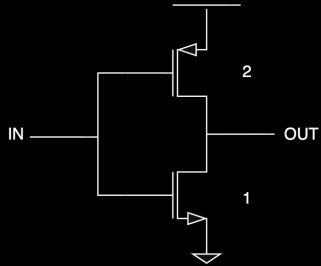


there are two pmos in this path
so we scale up by $2\times$ the pmos
inverter size!

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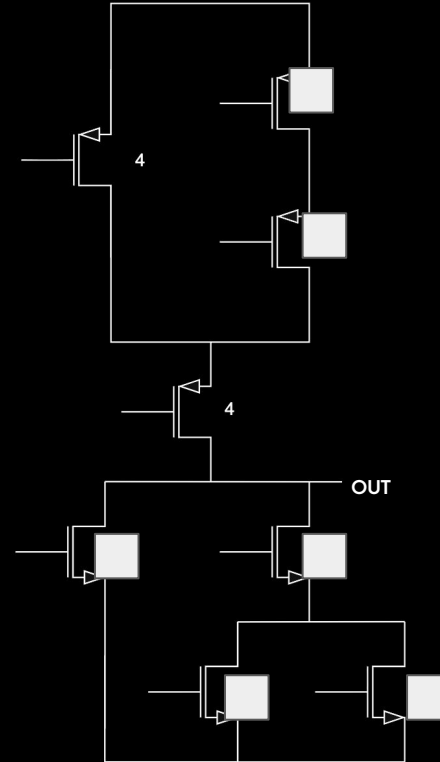


transistor sizing

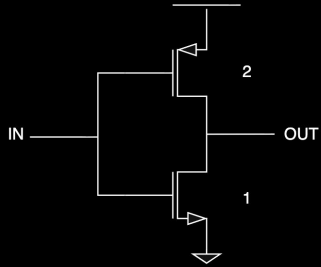


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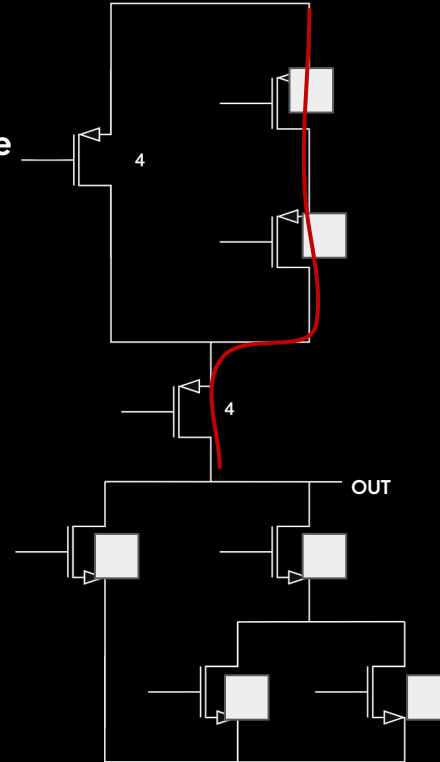


transistor sizing

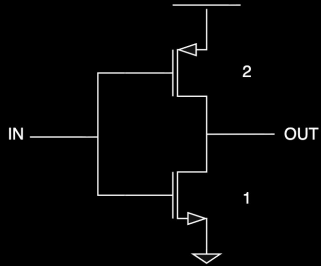


there are 2 pmos in this path
so we scale up by $2\times$ the pmos on the left!

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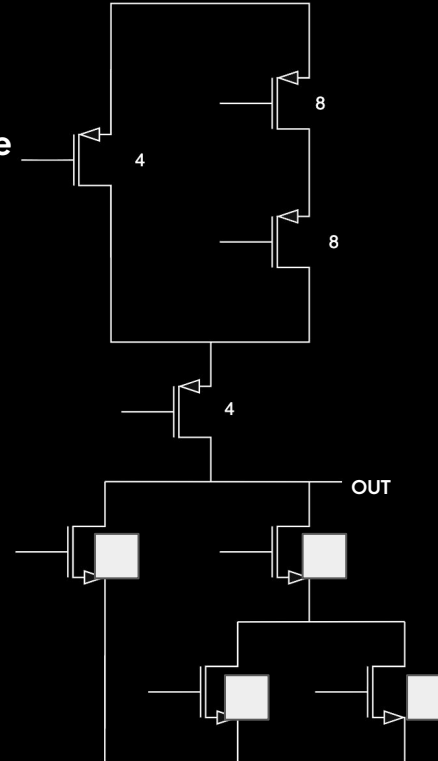


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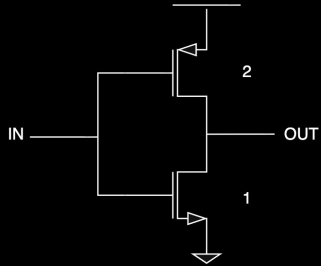


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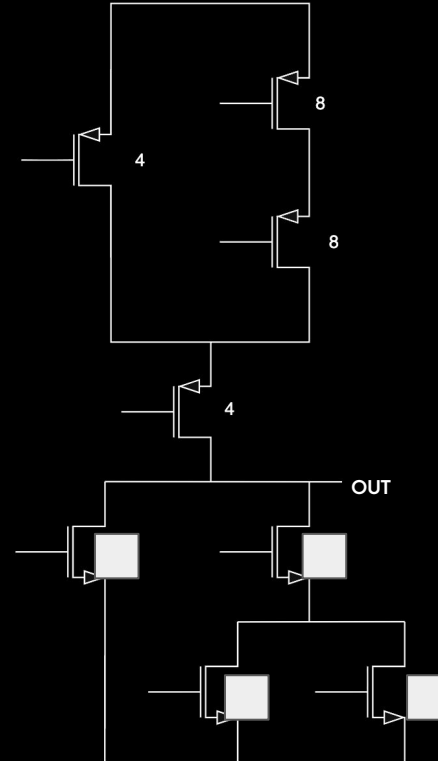


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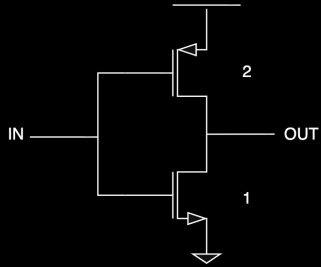


left or right?

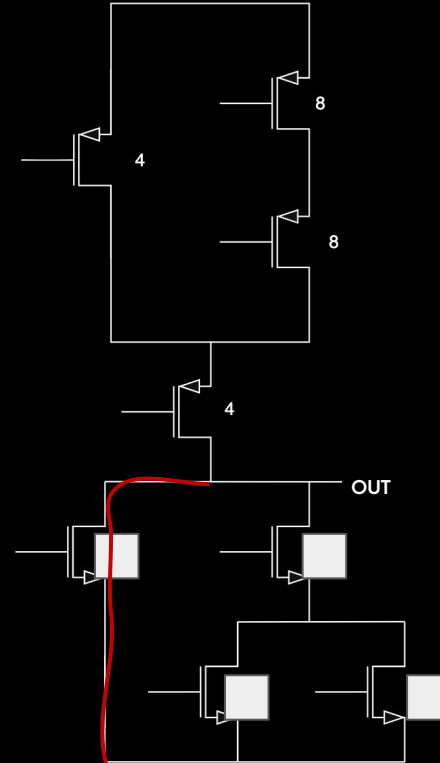
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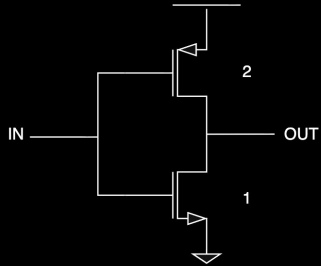
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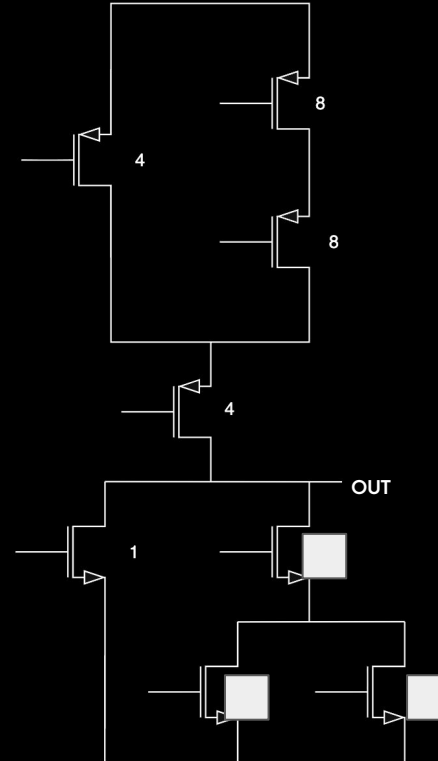


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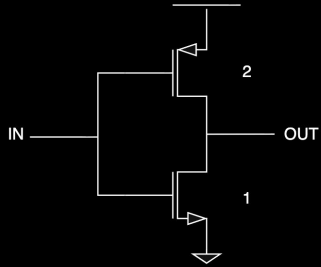


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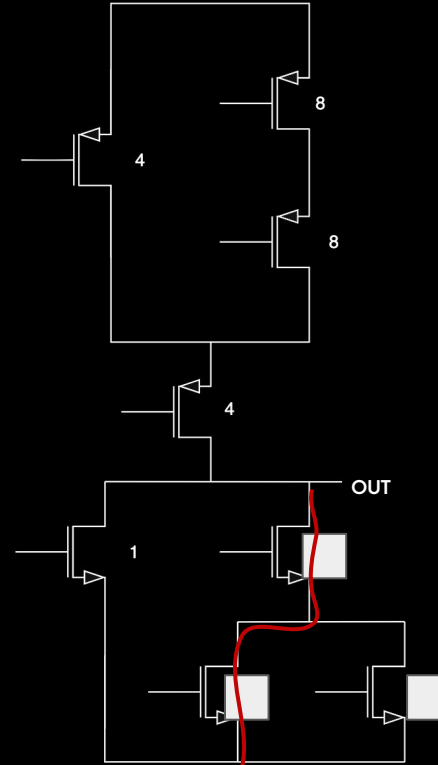


transistor sizing

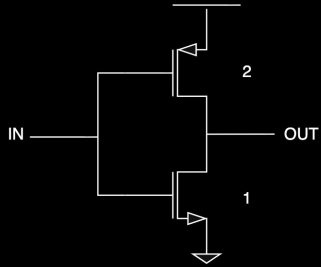


doesn't matter! both
are the same

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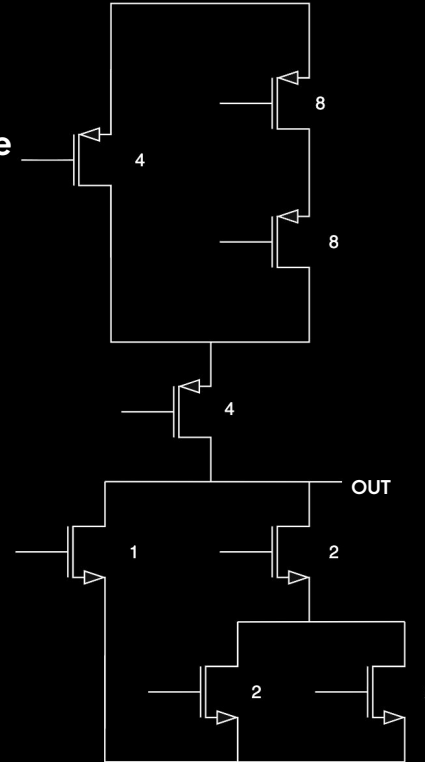


transistor sizing



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transistor sizing

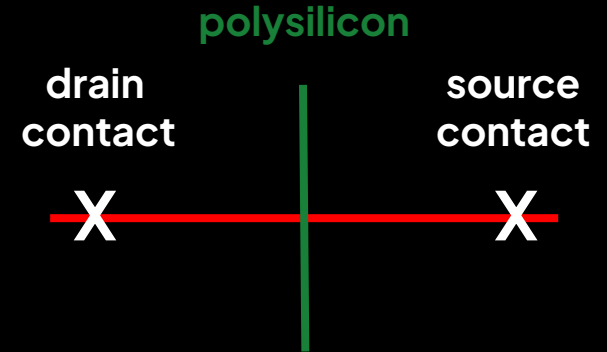
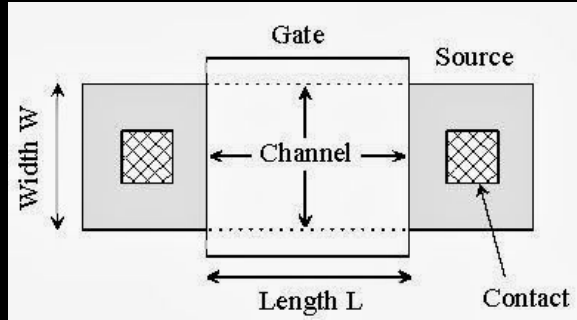


many **other** factors to consider:

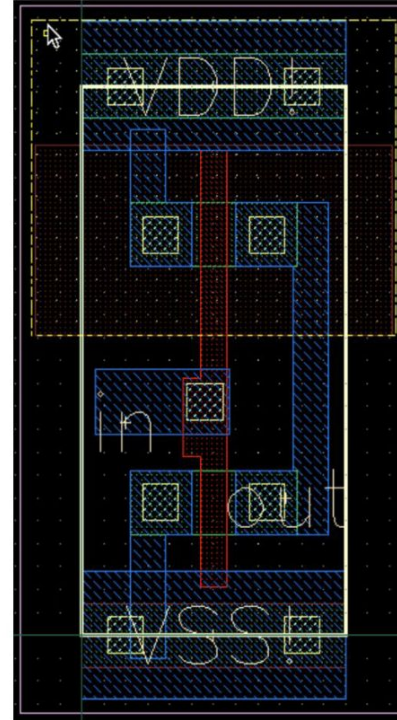
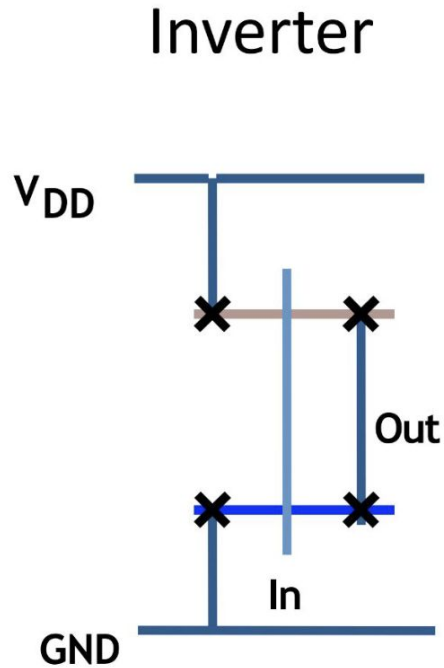
capacitance
speed
area

can discuss more if people are
interested

stick diagrams



stick diagrams



stick diagrams



why do stick diagrams?

enables you to map out
your layout beforehand!
(saves some headaches)

how to read DRC/LVS (custom)

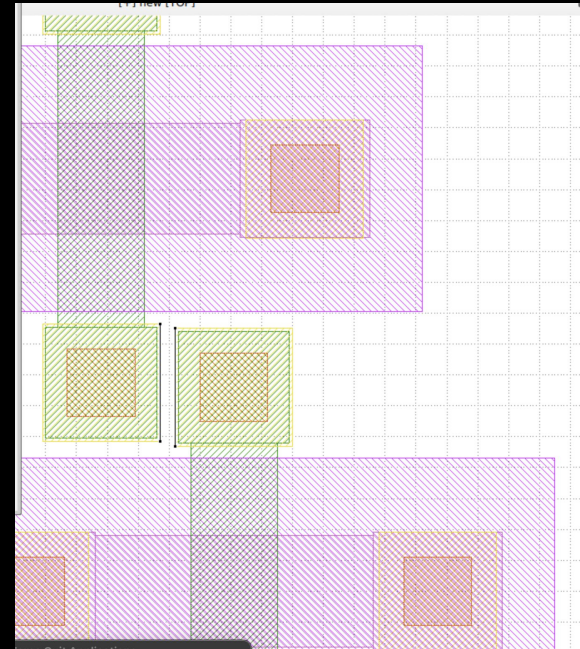
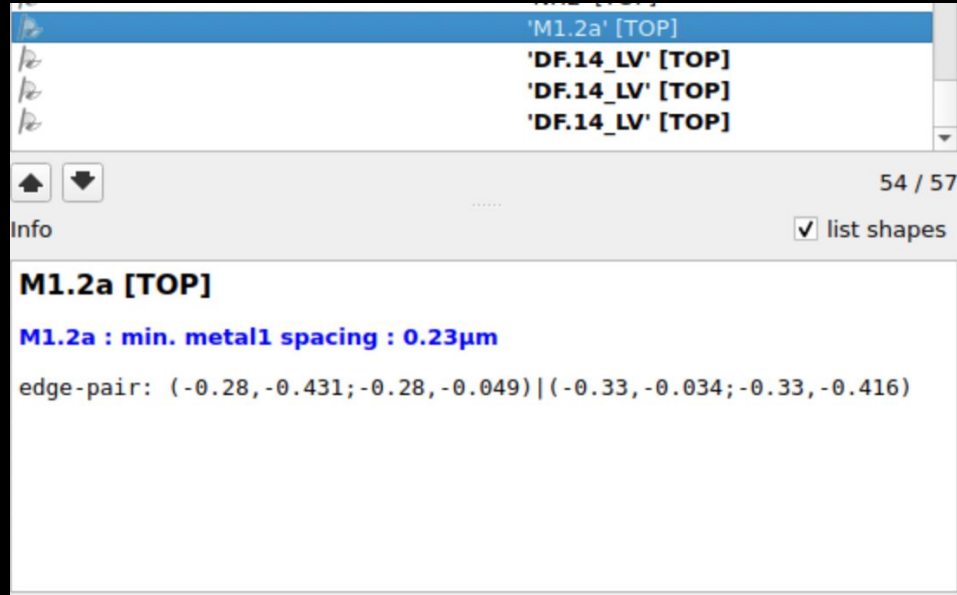


DRC: Design Rule Check
rules about your overall layout,
how minimum distance apart,
the minimum sizes, overlap, etc

LVS: Layout Vs Schematic
checking if your layout matches
your schematic

[full design manual](#)

how to read DRC/LVS (custom)



[full design manual](#)

how to read DRC/LVS (custom)



Mn.2a	Space	0.23 (n = 1) 0.28 (2 ≤ n ≤ 5)
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**minimum space
between metal N**

[full design manual](#)

how to read DRC/LVS (custom)



cheat sheet for DRC

full design manual

how to read DRC/LVS (custom)



Objects	Layout	Reference
▼  inv ↔ INV	inv	INV
▼ -> Pins		
▶ -> IN	IN (3)	IN (3)
▶ -> OUT	OUT (3)	OUT (3)
▶ -> VDD	VDD (3)	VDD (3)
▶ -> VSS	VSS (3)	VSS (3)
▶ ↑ Nets		
▶ ⌋ Devices		

what your layout sees

what your schematic sees

how to read DRC/LVS (custom)



use the trace tool to trace nets
Tools -> Trace Net, then click on your net

ensure that your layout matches
EXACTLY your schematic, down to the
transistor

be methodical



thank you for coming!



reading documentation after thai meeting